

WORK PLAN

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§1. INTRODUCTION: ADVANCING THE THEORY OF DISCRETE STRUCTURES

This document outlines a work plan for my role at the UPC, detailing strategies for contributing to both research and teaching in discrete mathematics.

Extremal and probabilistic combinatorics is a central and rapidly growing area in discrete mathematics, characterised by exciting recent advances and deep connections to theoretical computer science, number theory, and ergodic theory. At the same time, its accessibility at multiple levels makes it particularly valuable for teaching: introductory courses make use of simple, intuitive structures to train students in rigorous formal thinking, while advanced courses will equip them with the analytical tools needed to understand complex discrete networks—vital skills for today’s mathematicians and computer scientists.

UPC provides an ideal academic environment for this work, particularly given the renowned *Geometric, Algebraic and Probabilistic Combinatorics group* (GAPCOMB) and its vibrant community of scholars. This plan presents first a *research agenda* with both short-term objectives and a long-term vision for advancing the field (see Section 2). It also details a structured approach to teaching and student supervision, including specific proposals for teaching and mentoring at the Bachelor’s, Master’s, and PhD levels (see Section 4).

1.1. General introduction to Extremal and Probabilistic Combinatorics. In extremal (hyper)graph theory, the overarching objective is to establish how a (hyper)graph parameter forces the existence of certain rich substructure. This line of study can be traced back at least to Turán’s work [48] in the 1940s. While many extremal problems for *graphs* are relatively well understood, for hypergraphs our knowledge is still very limited. In the last two decades, researchers have been studying extremal questions for hypergraphs under various quasirandom constraints. This line of research has contributed to solving longstanding combinatorial questions and produced techniques applicable in broader mathematical contexts. For example, *the existence of designs* is one of the oldest problems in combinatorics and it was solved using methods that rely on extremal properties of quasirandom hypergraphs [28–30]. Furthermore, techniques used in this context, such as the well-known *Szemerédi’s regularity lemma*, have been further generalised, leading to developments in connection with other areas of mathematics.

Here we consider *quasirandom* hypergraphs. Roughly speaking, a graph or hypergraph is quasirandom if it shares some of the key properties of randomly generated graphs or hypergraphs. A property frequently used for this definition is having a uniform distribution of the edges. However, other ‘key properties’ of random graphs or hypergraphs could be considered instead. Interestingly, for graphs, many such properties yield essentially the same notion of quasirandomness (see the classical work by Chung, Graham, and Wilson [9]). In contrast, for hypergraphs, the discussion is much richer, where many non-equivalent notions of quasirandomness are studied.

Our core ambition is to advance our knowledge of hypergraph extremal problems under different constraints, particularly those involving quasirandom-type restrictions, while bringing these insights into the classroom to shape the formation of the next generation of mathematicians.

Brief comment on notation. We work primarily with *uniform hypergraphs*. A k -uniform hypergraph or k -graph H , is a pair (V, E) , where V is a (finite) set whose elements are called *vertices* and E is a set whose elements are k -element subsets of V , called *edges*. For $k = 2$, we will refer to them simply as *graphs*. The *complete k -graph* on n vertices contains all $\binom{n}{k}$ edges and is denoted by $K_n^{(k)}$.

We will consider hypergraphs with high *codegree*. For a k -graph H and a $(k - 1)$ -set $S \subseteq V(H)$ the *codegree* of S denoted $d(S)$ is the number of edges in H containing S . Moreover, $\delta(H)$ will denote the minimum codegree among all possible $(k - 1)$ -sets $S \subseteq V(H)$.

§2. RESEARCH AGENDA: SPECIFIC LINES OF DEVELOPMENT

The research agenda presented below focuses on two lines of research, which contextualise some of my most important results, while projecting new directions for future investigations. These lines of research, and the problems and conjectures in each case, are not exhaustive of the potential projects to be engage during my work at UPC. My work in extremal and probabilistic combinatorics is versatile and include various finite structures, including graphs [1, 2, 36], hypergraphs [3, 7, 10, 34, 37, 38, 40], and families of sets [20, 27, 41].

2.1. Turán-type problems for hypergraphs and its variations. One fundamental parameter in extremal graph theory is the so called *extremal number*, defined as follows. Given a graph G and an integer n

$$\text{ex}(n, G) = \max\{|E(H)| : H \text{ is an } n\text{-vertex graph not containing } G\}.$$

In a classical result, Erdős and Stone find a formula that relates the extremal number of any graph G with its *chromatic number* $\chi(G)$ *.

*For a graph G , its chromatic number $\chi(G)$ is the minimum number of colours needed to colour the vertices of G such that no edge is monochromatic.

Theorem 2.1. *For every graph G we have $\text{ex}(n, G) = \frac{\chi(G)-2}{\chi(G)-1} \binom{n}{2} + o(n^2)$.*

This surprisingly simple formula, asymptotically determines the edge density required to force the existence of any G as a sub graph of a sufficiently large H .

We may extend this definition to k -uniform hypergraphs instead of just graphs. Interestingly, Erdős-Stone theorem is not valid any more, and in fact, our knowledge about the behaviour of this fundamental parameter for hypergraphs remains still extremely limited. This, despite the significant attention gather from senior researches in the last 80 years, since the first results and questions started to emerge.

For now on we focus on 3-uniform hypergraphs. For a cleaner parameter, instead of the extremal number, academics have centred their attention on the study of the so-called *Turán density*. Given a 3-graph F , the *Turán density* of F , $\pi(F)$, is a normalised version of the extremal number given by the following limit

$$\pi(F) := \lim_{n \rightarrow \infty} \frac{\text{ex}(n, F)}{\binom{n}{3}}. \quad (2.1)$$

This limit exists for every 3-graph F since the sequence $\text{ex}(n, F)/\binom{n}{3} \geq 0$ is non-increasing.

While Theorem 2.1 and various related results were obtained for graphs, for hypergraphs this problem is much more challenging. In fact, determining the Turán density of $K_4^{(3)}$, the complete 3-graph on 4 vertices, is one of the most important open problems in the field and was already posed by Turán more than eighty years ago [48]. Even the seemingly easier problem of determining $\pi(K_4^{(3)-})$ is open as well, where $K_4^{(3)-}$ is the (unique) 3-graph on 4 vertices and 3 edges.

Conjecture 2.2. $\pi(K_4^{(3)}) = \frac{5}{9}$ and $\pi(K_4^{(3)-}) = \frac{2}{7}$

Since studying the Turán density $\pi(\cdot)$ turns out to be a very challenging problem, several variations and restrictions of this problem have been considered.

Codegree Turán density. Mubayi and Zhao [35] introduced the following variation of the extremal number. The *extremal codegree*, $\text{ex}_{\text{co}}(n, F)$, is the maximum $\delta(H)$ among all n -vertex F -free 3-graphs H . Moreover the *codegree Turán density* of F denoted by $\gamma(F)$ is given by the limit $\gamma(F) = \lim_{n \rightarrow \infty} \text{ex}_{\text{co}}(n, F)/n$ (and this limit always exists [35]). It is easy to see that $\gamma(F) \leq \pi(F)$ for every 3-graph F .

In a classical paper, Erdős [16] showed that for every 3-graph F we have $\pi(F) = 0$ or $\pi(F) \geq 2/9$. In contrast, Schülke and I [40] showed that the codegree Turán density has a very different behaviour around zero.

Theorem 2.3. *For every $\xi > 0$ there is a 3-graph F such that $0 < \gamma(F) < \xi$.*

In fact, we prove a general version of Theorem 2.3 for k -graphs with $k \geq 4$. Erdős's result mentioned above also holds for all $k \geq 4$ and $2/9$ replaced by $k!/k^k$.

Let $\Pi^{(3)} = \{\pi(F) : F \text{ is a 3-graph}\}$ and $\Gamma^{(3)} = \{\gamma(F) : F \text{ is a 3-graph}\}$. Erdős's result above implies that $\Pi^{(3)} \cap (0, 2/9) = \emptyset$ while our result implies that zero is an accumulation point of $\Gamma^{(3)}$. Recently the proof-method used for Theorem 2.3 was extended to find infinitely many accumulation point in $\Gamma^{(3)}$. Studying the properties and structure of these sets has attracted the attention of many researchers [18, 19, 35, 42], and is certainly within the scope of our research plans for the following years. In particular, determining the topological properties of the set $\Gamma^{(3)}$ will certainly shed light on our general understanding of Turán problems for hypergraphs.

Determining the value of γ is a very hard problem in general, and it is only known for few non-trivial hypergraphs (see Table 1 in [5]). Whilst the (classical) Turán density of $K_4^{(3)-}$ has not been determined, a computer-assisted proof [17] gives the following theorem.

Theorem 2.4. $\gamma(K_4^{(3)-}) = \frac{1}{4}$.

For $K_4^{(3)}$ the codegree Turán density remains unknown and Czygrinow and Nagle [12] conjectured it should be $1/2$. In a recent survey, Balogh, Clemen, and Lidický [5] systematically present, and in some cases improve, the best known lower and upper bounds for the codegree Turán densities of some small 3-graphs (up to six vertices). In particular, their results imply that $\gamma(C_5^{(3)-}) \leq 0.136$ and $\gamma(C_{10}^{(3)}) \leq 0.3993$, where $C_\ell^{(k)}$ is the (tight) k -uniform cycle on ℓ vertices and $C_\ell^{(k)-}$ is the (unique) k -graph obtained after deleting one edge from $C_\ell^{(k)}$. In recent papers, together with co-author Sales and Schülke [37] and subsequently Sanhueza-Matamala and Schacht [39], we determined the codegree Turán densities of $C_5^{(3)-}$ and $C_{10}^{(3)}$.

Theorem 2.5. $\gamma(C_5^{(3)-}) = 0$ and $\gamma(C_{10}^{(3)}) = 1/3$.

The study of the codegree Turán density of hypergraphs remains an active area of research including recent papers like [14].

Techniques for determine the codegree Turán densities are developing at a fast pace. I plan to continue the collaborations with my co-authors for the development of these techniques in order to tackle classical unknown codegree Turán densities. These include the most urgent open case, the clique on four vertices $K_4^{(3)}$.

In the longer vision, we expect that the determination of the codegree Turán density for specific hypergraphs together with the analysis of the general behaviour of the density as a function, will help us determine an algorithm for the Turán density of any hypergraph F . Such algorithm will certainly be more complex than the elegant solution found by Erdős and Stone in Theorem 2.1, and it will require a subtle analysis of the hypergraphs involved.

As an intermediate steps for such algorithm we expect to find a solutions for complete hypergraphs first, which already offers a tremendous challenge. Another very important intermediate result would be to settle the following conjecture on possible codegree Turán densities.

Conjecture 2.6. $\mathbb{Q} \cap [0, 1) \subseteq \Gamma^{(3)}$.

Weakly and strongly uniform hypergraphs. Erdős and Sós [15] suggested to study Turán-type questions in quasirandom 3-graphs. Roughly speaking, a hypergraph is *weakly d -dense* (for some $d \in [0, 1]$) if every linearly sized set of vertices has edge-density at least d . Then, the *uniform Turán density* $\pi_{\bullet}(F)$ of a 3-graph F is essentially the minimum d such that every uniformly $(d + o(1))$ -dense 3-graph contains a copy of F . In contrast with the classical case, the uniform Turán density of $K_4^{(3)-}$ is now fully determined.

Theorem 2.7. $\pi_{\bullet}(K_4^{(3)-}) = 1/4$.

Theorem 2.7 was first proven by Glebov, Král', and Volec [23] through a computer-assisted proof and later by Reiher, Rödl, and Schacht [45] by an application of the *regularity method for hypergraphs*. Still, the value of $\pi_{\bullet}(K_4^{(3)})$ remains unknown.

In a series of papers (see [43] for a comprehensive survey), Reiher, Rödl, and Schacht went further and investigated Turán-type problems for other kinds of uniform density conditions. As systematically studied in [47] and in contrast with the graph case (see e.g. [9]), there are many non-equivalent notions of quasirandomness in hypergraphs. In each case there is a related Turán-type question. In [44], they studied the strongest of these quasirandom conditions for which the Turán problem is not yet trivial. For simplicity we omit the details of the definition of this type of density and we refer to it as the *strong uniform density condition*. The respective Turán parameter will be called the *strongly uniform Turán density*, and for every 3-graph F we denote it as $\pi_{\blacktriangle}(F)$.

Berger, Reiher, Rödl, Schacht, and I [7] determined $\pi_{\blacktriangle}(K_5^{(3)}) = 1/3$, where $K_5^{(3)}$ is the complete 3-graph on 5 vertices.

Theorem 2.8. $\pi_{\blacktriangle}(K_5^{(3)}) = \frac{1}{3}$.

Before Theorem 2.8, the Turán density of $K_k^{(3)}$ was obtained [44] for all $k \leq 16$ except for $k = 5, 9$, or 10 . Perhaps surprisingly, the case of $K_5^{(3)}$ remained as a gap in the theory until our paper, and ended up requiring a much more difficult and subtle proof than for other complete hypergraphs. Our proof is based on the *hypergraph regularity method* followed by a thorough analysis of the ‘holes’ in $K_5^{(3)}$ -free hypergraphs.

Studying Turán-type problems in different kinds of quasirandom host hypergraphs remains an active field of research and has now garnered wide attention in its own right (see e.g. the recent papers [8, 11, 21, 22, 32, 46]).

There is a specific open question in the area, that will serve us to project a longer term vision for the research on quasirandom Turán problems. The following question was posed by Schacht in [46] and it definitely belongs within the scope of our research plans for the following years.

Question 2.9. Is it true that $\pi_{\bullet}(F) < \pi(F)$ for every 3-graph with $\pi(F) > 0$?

Resolving this question would provide deep insights into how quasirandomness influences Turán-type problems, a theme that connects all the variations discussed in this section. More generally, we would want to understand the relation between the strong and the weak quasirandom variations versus hypergraphs with minimum codegree conditions, as studied in the previous subsection.

§3. DESIGNS AND DECOMPOSITIONS INTO SPANNING SUBGRAPHS

3.1. Decompositions of the complete hypergraph. One of the oldest results in combinatorics is due to Kirkman [31] who proved in 1847 that the edges of the complete graph on n vertices, K_n , can be partitioned into a collection of copies of K_3 for every odd n such that $\binom{n}{2}$ is a multiple of 3. It is easy to check that these divisibility conditions are necessary for such a partition to exist. For a more general presentation, we say that a *decomposition of a k -graph H* is a collection of subgraphs such that every edge of H is covered exactly once. When these subgraphs are all isomorphic to a single k -graph F we say that it is an *F -decomposition*. For such a decomposition to exist, some divisibility conditions are obviously necessary, e.g. the number of edges of H must be divisible by the number of edges of F . In general we will not specify these conditions and we will simply say that H is *F -divisible* when they are satisfied. In this framework, Kirkman's result above says that if K_n is K_3 -divisible, then it contains a K_3 -decomposition. Finding sufficient conditions for F -decompositions in F -divisible hypergraphs is an intensively studied problem.

In recent years several new decomposition results have been proven, including some major problems. In particular, Keevash [29] generalised Kirkman's result by proving that for any $\ell \in \mathbb{N}$ and a sufficiently large n , if $K_n^{(k)}$ is $K_\ell^{(k)}$ -divisible, then it contains a $K_\ell^{(k)}$ -decomposition. Moreover, he proved a multigraph version of this result that solved a problem posed by Steiner in the mid-19th century about the existence of (combinatorial) *designs*. Later, Glock, Kühn, Lo, and Osthus [25] generalised this result by showing that sufficiently large F -divisible complete hypergraphs contain F -decompositions for every F .

Theorem 3.1. *Let F be a k -graph and let $n \gg |V(F)|$ such that $K_n^{(k)}$ is F -divisible. Then $K_n^{(k)}$ contains an F -decomposition.*

A key restriction of Glock, Kühn, Lo, and Osthus's result is that it only applies when F is much smaller than the host complete hypergraph. In the opposite end, we might study the case in which F is a connected *spanning* hypergraph.

Recently we have been working on these problems with Richard Lang a member of GAMCOMB at UPC. In fact, together with Sanhueza-Matamala [33], we solved the first natural instance of this problem by showing the following result for (tight) cycles in k -uniform hypergraphs.

Theorem 3.2. *For every $k \geq 2$, there is n_0 such that for every $n \geq n_0$ dividing $\binom{n}{k}$, the complete k -uniform hypergraph of order n has a decomposition into tight Hamilton cycles.*

This result confirms a conjecture posed by Bailey and Stevens [4]. Moreover, our methods extend to other structures beyond Hamilton cycles, including the so-called ‘*wreaths*’, confirming an elegant conjecture posed by Baranyai [6] in 1975.

Our proof represents a significant departure from previous decomposition methods, as we developed new techniques to handle the *global connectivity* of Hamilton cycles. We plan to continue developing these methods to tackle the open problems that remain in the area, including the hypergraph generalization of the so-called Oberwolfach problem. The graph version of this problem was famously solved asymptotically by Glock, Joos, Kim, Kühn, and Osthus [24], who also suggested several variations asking for decompositions into connected spanning subgraphs. We believe our method can be adapted to address many of these problems. However, our techniques must be tailored to each case, and a general solution remains out of reach. Such a generalization would be a major challenge, requiring rigorous systematization of our methods—a central goal in our research plans for the coming years at UPC.

In the long term, a more unified approach should bring together decompositions into spanning hypergraphs with decompositions into smaller substructures, where our understanding is much more developed. This would yield shorter and more fundamental proofs than our current methods.

3.2. Decompositions in hypergraphs with minimum codegree conditions. Having established decomposition results for complete hypergraphs, a natural next direction is to study *sparser* hosts. In particular, we might start exploring decompositions in hypergraphs with minimum codegree conditions. Together with Sanhueza-Matamala, we showed the following result for cycles in 3-graphs.

Theorem 3.3. *Let $\ell \geq 10^7$ and let H be a 3-graph $C_\ell^{(3)}$ -divisible 3-graph on $n \gg \ell$ vertices. If $\delta(H) \geq (2/3 + o(1))n$, then H contains a $C_\ell^{(3)}$ -decomposition. Moreover, for every $n \geq \ell$ divisible by 18 there exists a 3-divisible 3-graph F with $\delta(F) \geq 2n/3 - 5$ which does not contain a $C_\ell^{(3)}$ -decomposition.*

This result remains the only known optimal codegree condition for decompositions in hypergraphs, apart from an older result by Yuster on hypertrees [50]. Theorem 3.3 disproved a conjecture by Glock, Kühn, and Osthus [26] who believed that codegree $(1/2 + o(1))n$ would be enough to prove the existence of such decompositions. Later, together with Sanhueza-Matamala and Lo [34] we generalised the first part of our result to k -graphs for every $k \geq 4$. More precisely, we showed that every sufficiently large $C_\ell^{(k)}$ -divisible k -graph H with $\delta(H) \geq (2/3 + o(1))|V(H)|$ contains a $C_\ell^{(k)}$ -decomposition. However, for $k \geq 4$ we do not know whether the constant ‘ $2/3$ ’ here is optimal or not. In fact, the aforementioned conjecture by Glock, Kühn, and Osthus [26] was posed for k -graphs, and it might well be that it only fails for $k = 3$.

Question 3.4. Let $k > 3$ and ℓ sufficiently large. Given $n_0 \gg \ell$, is it true that any $C_\ell^{(k)}$ -divisible k -graph H on $n \geq n_0$ vertices with $\delta(H) \geq (1/2 + o(1))n$ contains a $C_\ell^{(k)}$ -decomposition?

Crucially, in all the results above, we have $n \gg \ell$, i.e. the cycle is very small compared with the host hypergraph. As in Subsection 3.1 we are interested in finding optimal codegree conditions for decompositions into *spanning* hypergraphs. The following conjecture was posed by Glock, Kühn, and Osthus and would be the first step in this direction.

Conjecture 3.5. Let $k \in \mathbb{N}$ and suppose n is sufficiently large and divisible by k . Let H be an n -vertex k -graph with $\delta(H) \geq (1/2 + o(1))n$ and such that every vertex lies in the same number of edges. Then H is decomposable into perfect matchings.

If true, this conjecture would be optimal and would generalise a classical result by Baranyai [6] which establishes the existence of such decompositions in the complete hypergraph.

We finish this section with the following very challenging conjecture.

Conjecture 3.6. Let $k \in \mathbb{N}$ and suppose n is sufficiently large and divides $\binom{n}{k}$. Let H be an n -vertex k -graph with $\delta(H) \geq (2/3 + o(1))n$ and such that all vertices lie in the same number of edges, which is a multiple of k . Then H is decomposable into Hamilton cycles.

The resolution of this conjecture would represent a landmark achievement. It brings together the difficulties found in decomposition into spanning while operating under asymptotically optimal codegree conditions.

§4. TEACHING AND SUPERVISION

Teaching represents one of the fundamental dimensions of our contribution as a mathematicians—the transmission of knowledge and the cultivation of rigorous thinking. My commitment to this mission began during my undergraduate and Master’s studies, where I

actively pursue teaching opportunities. Following a research-intensive PhD, my postdoctoral position at the University of Birmingham allowed me to bring my research expertise to the classroom, teaching advanced topics in combinatorics. This experience reinforced my belief that impactful teaching and cutting-edge research strengthen one another: research informs pedagogy with current developments, while teaching clarifies and redefines theoretical understanding.

4.1. Previous experience. My teaching experience began at the *Universidad de Chile*, where advanced Bachelor's and Master's students commonly assume significant teaching responsibilities. As a teaching assistant, my role extended beyond leading exercise sessions and grading. I was frequently responsible for delivering the core theoretical content for fundamental undergraduate courses in mathematics and computer science, including *Calculus*, *Algebra*, *Discrete Mathematics*, *Graph Theory*, and *Databases*. This provided deep, practical experience in curriculum delivery.

Moreover, I participated in the pedagogical innovation project "*Los Dos Relojes*," which extended beyond expository lectures to provide direct support for students struggling with abstract concepts through alternative teaching approaches. The project included interdisciplinary activities that integrated mathematics with other fields. This experience helped me develop a practical toolkit for adaptive and supportive teaching methodologies that complement the traditional lecture-based instruction.

At the *University of Birmingham*, I taught in a research-intensive, English-language environment. I taught a master's-level course on *Advanced Topics in Combinatorics* and led the reading seminar on *Extremal Set Theory* for PhD students. I also co-supervised a *Master's student* through their dissertation on "*Extremal Problems for Posets*," which involved weekly guidance on bibliographical research and technical writing.

Looking ahead, I am engaged to a full academic integration at UPC and in Catalonia. Having taught in both Spanish and English, I am looking forward to undertaking a formal Catalan language course to expand my teaching capacity and connect more teaching opportunities in Catalonia.

4.2. Example course: *The Probabilistic method*. As part of this work plan, I want to provide a concrete illustration of my teaching vision. Below, I present some details of an advanced course in extremal combinatorics that draws on my research expertise. That said, I am equally prepared and enthusiastic about teaching fundamental courses across mathematics, computer science, and engineering programs.

Context and motivation. The *Probabilistic Method* is one of the foundations of contemporary combinatorics, based on a profound original insight by Paul Erdős. This method uses probability theory to prove the existence of combinatorial structures, often leading

to elegant and surprising solutions to fully deterministic problems. For students, it helps develop mathematical intuition, demonstrating how randomness can help in understanding seemingly chaotic and complex discrete systems.

The journey from basic applications—Erdős’s original idea for a lower bound for diagonal Ramsey numbers—to advanced techniques involving more subtle concentration inequalities reveals a tremendously rich theory. This progression will deepen students’ analytical skills and develop the intuition required for understanding intricate networks, randomised algorithms, and large-scale data structures, which are fundamental skills for mathematicians and computer scientists nowadays.

This subject naturally connects probability, linear algebra, and graph theory within a concrete, problem-solving framework. Its emphasis on elegant non-constructive arguments is a significant intellectual challenge and a potential inspiration for advanced Bachelor and Master’s students.

Basic syllabus. The course is structured into three core modules.

Module 1: *Foundational Probabilistic Arguments.*

This module establishes the core philosophy of the method. Students will learn to use basic probability theory to prove deterministic existence theorems.

Key Topics: *Basic ideas. Sampling. Alterations. Markov Inequality. Second moment method.*

Module 2: *Advanced Concentration Inequalities and Martingales.*

This module covers more complex tools to understand probabilistic constructions sequentially, and controlling deviations in random structures. These are essential to solve natural concrete extremal problems.

Key Topics: *Chernoff bound. Azuma-Hoeffding bound. McDiarmid’s inequality. Applications: Discrepancy, Clique and Chromatic numbers.*

Module 3: Local Lemma and Dependent Random Choice.

The final module explores these last two techniques with their particular applications. These are simple probabilistic techniques designed specifically to tackle combinatorial problems. In this way, the course closes by returning to the core idea of the method.

Key Topics: *Local Lemma, Dependent Random choice, Applications: Hypergraphs colouring, Latin squares, Turán numbers for bipartite graphs, Ramsey number of bounded degree graphs.*

Classes. Theoretical lectures for each module will establish rigorous definitions and proofs, organized around the core topics of the course. These sessions will be interactive, encouraging students to construct examples and engage with the intuitions behind formal proofs. Problem sessions will run concurrently to explore further applications utilising

probabilistic constructions, most of the times, similar to those introduced in lectures. On particular occasions, novel applications will also be presented during problem sessions, exposing students to innovative challenging problems that extend beyond the standard material.

To support lectures, the course material will be made available online from the outset—lecture notes, problem sheets, and bibliographical references. Regular office hours will provide an essential forum for students to address their questions directly, discuss course topics in depth, and receive individualized guidance on challenging material.

4.3. Graduate and advisory roles. Mentoring students through research projects is one of the central aspects of academic life. My successful experience co-supervising a Master’s thesis at Birmingham—where the student deepened her knowledge, mastered diverse mathematical tools, and developed the crucial intuition for tackling complex discrete problems—confirmed the profound impact of effective guidance.

Below, I outline my approach to supervision at different academic levels, along with very brief descriptions of potential project ideas I would be eager to develop with students at UPC.

Bachelor’s Thesis The work for a bachelor’s thesis should be intellectually challenging yet avoid overly technical machinery. It should centre on a single, contemporary idea, allowing the student to engage directly with cutting-edge developments in mathematics. A well-executed Bachelor’s thesis should involve understanding and reformulating this contemporary idea, which can naturally serve as a foundation for a subsequent Master’s thesis.

The following are potential projects to develop with students.

Understanding the examples for clique decompositions in hypergraphs. Constructing hypergraphs with high minimum codegree that avoid clique decompositions is a complex task, requiring a synthesis of many combinatorial ideas. Recent developments have yielded novel constructions (see [13]). A bachelor’s thesis could provide an excellent opportunity to achieve a detailed understanding of these constructions and present a clear, verified proof of their non-decomposability.

Ramsey numbers of G^- and G^+ . This project would investigate how small alterations — such as adding or removing an edge or vertex — affect the Ramsey number of a graph. The core work involves a thorough literature review, followed by the systematization and comparative analysis of existing results and proof techniques. Progress on this question is scattered across multiple papers, including a recent and elegant contribution by Wigderson [49]. Given its scope, successful work on this survey could naturally extend into a Master’s thesis.

Master's thesis. The core strength of a Master's thesis should be the adaptation of known techniques to new problems, requiring students to deeply understand these techniques and their limitations. This process is fundamental for developing mature research judgment and gaining the capacity to analyse complex discrete structures. Although they may be more technical than a Bachelor's thesis, a Master's thesis should avoid convoluted technicalities and instead focus on clear topics where relatively simple proof strategies can be applied effectively.

The following are potential projects to develop with students.

Hamilton Cycles. There has been many successful applications of absorption for different notions of cycles in hypergraphs. However, the notion of *Cycle* has not been yet studied. Finding (co)degree conditions for the existence of a Hamilton Cycles is an interesting open question for a master student.

Find a 3-graph with codegree Turán density $1/3$. The set of possible codegree Turán densities for a single 3-graph has been increasingly studied in recent years. However, finding a 3-graph whose codegree Turán density is significantly larger than zero but smaller than $1/2$ remains elusive. Numerous techniques and candidate hypergraphs exist that could satisfy this property.

This project has the potential to evolve into part of a Ph.D. project, depending on the depth of the results obtained.

Ph.D. thesis. A successful Ph.D. project should be ambitious and feasible, beginning with accessible problems that build the student's confidence and technical skills before progressing to more challenging questions. Throughout this progression, problems should remain genuinely demanding and lead to novel contributions to the research area. Many of the problems discussed in Section 2 are suitable starting points for such a trajectory.

Below, we exemplify problems that represent natural next steps from current results and methods. While solving them will likely require significant modification of existing techniques, they appear to remain within the scope of current methodological frameworks. Success in any of these directions would naturally open up new avenues for discussion and further research.

Stability argument for iterative palette patterns. The study of possible weakly Turán densities has changed dramatically in recent years, due to significant new results. However, the existence of extremal examples that arise from iterative patterns remains an open problem. We believe that a careful analysis of current techniques could lead to such a result. Several smaller subproblems need to be addressed along the way, though.

Jumps in the uniformly dense Turán. While the new results mentioned in the previous project show that non-jumps exist for the weak quasirandom Turán density, we have not

determined the existence of *jumps* (except for the one taking place at zero). Further analysis of the current techniques should clarify whether jumps exist, but significant modifications of the current techniques are needed.

4.4. Advanced Seminars and Reading Groups. I am keen to organise and participate in reading seminars. I believe they serve two important purposes: first, they keep researchers updated on new techniques and foster communication with scholars from other areas and institutions (as the presenter need not be the paper’s author). Second, they provide a valuable formative experience for students, who must prepare talks and extract the core arguments within a limited time.

My approach is informed by practice at the University of Birmingham, where it is customary for postdocs to organise a reading seminar for PhD students. It takes place every semester and is focused on a specific topic. This format allows for a deep dive into a subject while significantly broadening the Ph.D. students’ foundational knowledge.

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